
2026 Chantilly Math Competition

Elementary School Division - Individual Round Solutions

1. (9 pts) What is the mean of the numbers 15, 30, 45, 60, and 75?

Madhavan Kandagatla

Solution

$$\frac{15 + 30 + 45 + 60 + 75}{5} = \boxed{45}.$$

2. (9 pts) Nischal takes a class with 7 tests. Nischal's average score on the first 6 tests is 76. If Nischal needs his average score on all 7 tests to be at least 67 to pass, what is the minimum possible score Nischal could get on his 7th test and still pass?

Aryan Raj

Solution

$$7 \cdot 67 - 6 \cdot 76 = \boxed{13}.$$

3. (10 pts) Which integer is the closest to the circumference of a circle with radius 9?

Rudra Khanwalkar

Solution

$$2 \cdot \pi \cdot 9 = 18\pi \approx 56.55$$

so the answer is $\boxed{57}$.

4. (10 pts) Farmer John only has chickens and cows on his farm. There are 67 animals on the farm. These animals have 200 legs altogether. How many chickens are on the farm?

Aryan Raj

Solution

$$\frac{67 \cdot 4 - 200}{4 - 2} = \boxed{34}.$$

5. (11 pts) A bag has 6 red balls, 7 green balls, and 8 blue balls. What is the fewest number of balls that need to be removed from the bag so that there are more red balls than balls that aren't red?

Aryan Raj

Solution

$$8 + 7 - 6 + 1 = \boxed{10}.$$

6. (11 pts) Boris is buying a sweater. The price tag says \$100 on it. Boris has a coupon that gives him a 6% discount, but there is also a 7% sales tax. After the coupon and sales tax are both applied, by how many cents does the price change?

Aryan Raj

Solution

$$100(100 \cdot .94 \cdot 1.07 - 100) = \boxed{58}.$$

7. (12 pts) A plane is scheduled to depart from City A at 5 PM and arrive at City B at 8 PM, traveling at an average speed of 550 mph. If the plane is instead delayed by 30 minutes, what is the new average speed required to arrive at City B on time, in mph?

Haorui Zhao

Solution Since it takes $8 - 5 = 3$ hours to go from City A to City B at an average speed of 550 mph, the distance between the cities is $550 \cdot 3 = 1650$ mi. When the plane is delayed by 30 minutes, it now has $\frac{5}{2}$ hours to travel the same distance to arrive on time, so its new average speed has to be $\frac{1650}{\frac{5}{2}} = \boxed{660}$ mph.

8. (12 pts) Darren has a spherical soap. After 234 showers, the soap is a sphere with 40% of its original radius. Assuming every shower uses the same amount of soap, how many more showers will the soap last?

Aryan Raj

Solution

$$\frac{.4^3}{1 - .4^3} \cdot 234 = \boxed{16}.$$

9. (13 pts) Find the smallest odd prime that is 2 more than a multiple of the sum of its digits.

Aryan Raj

Solution Clearly, this prime can't be 1 digit.

When it has 2 digits, let it be $10a + b$, so

$$-9b \equiv 10a + b \equiv 2 \pmod{a + b}.$$

Therefore, $a + b$ must be a divisor of $9b + 2$.

Since $10a + b$ is prime, we must have $b \in \{1, 3, 7, 9\}$.

When $b = 1$, $9b + 2 = 11$, so there is no a .

When $b = 3$, $9b + 2 = 29$, so there is no a .

When $b = 7$, $9b + 2 = 65$, so $a = 13 - 7 = 6$ is the smallest a that works.

When $b = 9$, $9b + 2 = 83$, so there is no a .

Therefore, the answer is $10(6) + 7 = \boxed{67}$.

10. (13 pts) Alice writes the positive integers from 1 to n inclusive on a whiteboard. Bob erases some (but not all) of the positive integers, so the average of the remaining integers is exactly 6.7. What is the minimum possible value of n ?

Aryan Raj

Solution The remaining integers sum to an integer, and $6.7 = \frac{67}{10}$, so $n > 10$.

When $n = 11$, the total sum is $\frac{11 \cdot 12}{2} = 66$, which is too small.

When $n = 12$, the total sum is $\frac{12 \cdot 13}{2} = 78$, so Bob can erase 5 and 6.

Therefore, the answer is $\boxed{12}$.

11. (14 pts) Compute the smallest positive integer divisible by 999 with all digits differing.

Aryan Raj

Solution Since $999k = (1000 - 1)k$, the last 3 digits of $999k$ will be $1000 - k$.

The largest positive integer less than 1000 with distinct digits is 987.

Since $1000 - k = 987$, $k = 13$, so the answer is $999k = 999(13) = \boxed{12987}$.

12. (14 pts) If

$$\frac{20}{23} = \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}}$$

for positive integers a , b , c , and d , compute $1000a + 100b + 10c + d$.

Aryan Raj

Solution

$$\frac{20}{23} = \frac{1}{1 + \frac{1}{6 + \frac{1}{1 + \frac{1}{2}}}},$$

so $1000a + 100b + 10c + d = 1000(1) + 100(6) + 10(1) + (2) = \boxed{1612}$.

13. (15 pts) Fill in the blanks with the digits 1, 2, 3, and 4 such that no two digits in the same row, column, or 2×2 box are the same.

1	2		-	-
-	-		-	-
-	-		-	-
3	-		-	4

Let the digits in the third column from the left be a, b, c , and d from top to bottom.

Compute $125a + 25b + 5c + d$.

Aryan Raj

Solution

1	2		4	3
4	3		1	2
2	4		3	1
3	1		2	4

Therefore, the answer is $125(4) + 25(1) + 5(3) + (2) = \boxed{542}$.

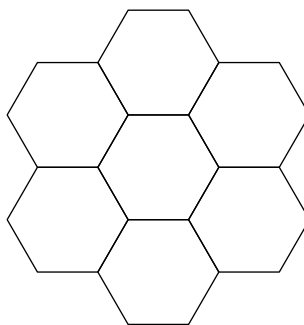
14. (15 pts) A circle has area 10. Another shape has twice the perimeter of the circle. What is the smallest positive integer that can't be the area of this other shape?

Aryan Raj

Solution

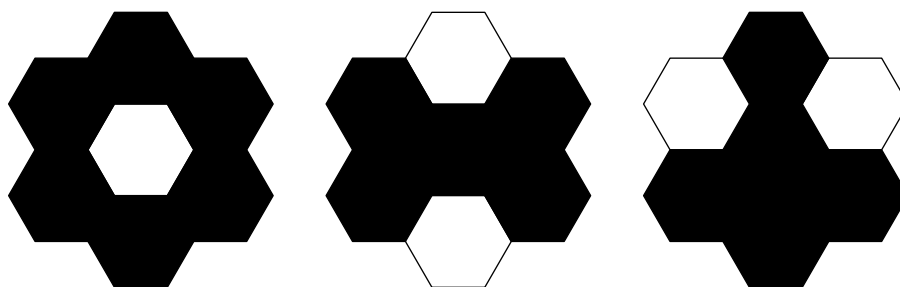
$$10 \cdot 2^2 + 1 = \boxed{41}.$$

15. (16 pts) How many distinct subsets of the 7 hexagons can be colored black such that there are exactly 6 unordered pairs of adjacent black hexagons?



Aryan Raj

Solution There are 3 cases.



Therefore, the answer is $1 + 3 + 6 = \boxed{10}$.

16. (16 pts) Call a positive integer steven if the sum of its digits is even. Compute the 2026th smallest steven positive integer.

Aryan Raj

Solution

$$2026 \cdot 2 + 1 = \boxed{4053}.$$

- TB. (Tiebreak) Shubham loves chess, but only has the black pieces. Therefore, he has a chessboard, which is a rectangular grid with 8 squares on each side, and 16 black chess pieces. Assuming all black chess pieces are indistinguishable and each of the 64 squares on the chessboard has at most 1 piece, estimate the number of ways Shubham can place the 16 black chess pieces on the chessboard such that each of the 8 rows and each of the 8 columns has exactly 2 black pieces in it.

Aryan Raj

Solution We biject each way to a graph with 16 vertices $r_1, c_1, r_2, c_2, \dots, r_8$, and c_8 by drawing an edge between r_i and c_j iff there is a chess piece in the i th row and j th column.

Since each row and each column has 2 chess pieces in it, all 16 vertices have degree 2.

This means that the graph can be decomposed into cycles.

Since the graph is bipartite, each of these cycles must have even length.

Let there be k_l cycles of length $2l$, where $l \geq 2$.

Then, there are

$$\frac{(8!)^2}{\prod_{i=2}^{\infty} ((2i)^{k_i} k_i!)}$$

distinct graphs resulting from labelling the vertices.

Since $\sum_{i=2}^{\infty} 2ik_i = 16$, we have $\sum_{i=2}^{\infty} ik_i = 8$, so the only cases are:

- (a) $k_8 = 1$
- (b) $k_6 = 1, k_2 = 1$
- (c) $k_5 = 1, k_3 = 1$
- (d) $k_4 = 2$
- (e) $k_4 = 1, k_2 = 2$
- (f) $k_3 = 2, k_2 = 1$
- (g) $k_2 = 4$

Therefore, the number of ways is

$$\frac{(8!)^2}{16} + \frac{(8!)^2}{12 \cdot 4} + \frac{(8!)^2}{10 \cdot 6} + \frac{(8!)^2}{8^2 \cdot 2!} + \frac{(8!)^2}{8 \cdot 4^2 \cdot 2!} + \frac{(8!)^2}{6^2 \cdot 2! \cdot 4} + \frac{(8!)^2}{4^4 \cdot 4!} = \boxed{187530840}.$$